AS · Cambridge (CIE) · Physics

17 mins 17 questions

Multiple Choice Questions

# Physical Quantities

Physical Quantities

| | |
|---|---|
| Easy (6 questions) | /6 |
| Medium (5 questions) | /5 |
| Hard (6 questions) | /6 |
| **Total Marks** | **/17** |

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# Easy Questions

**1** Which definition is correct and uses only quantities rather than units?

**A.** Density is mass per cubic metre

**B.** Potential difference is energy per unit current

**C.** Pressure is force per unit area

**D.** Speed is distance travelled per second

**(1 mark)**

**2** What is a reasonable estimate for the gravitational force acting on an adult person standing on the Earth?

**A.** 50 N

**B.** 300 N

**C.** 400 N

**D.** 700 N

**(1 mark)**

**3** A physical quantity has the symbol $V$ and the unit $m^3$.

Which physical quantity does $V$ represent?

**A.** Potential difference

**B.** Velocity

**C.** Pressure

**D.** Volume

**(1 mark)**


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**4** What is a reasonable estimate of the number of seconds in a year?

**A.** $5 \times 10^5$ s

**B.** $1 \times 10^6$ s

**C.** $3 \times 10^7$ s

**D.** $7 \times 10^8$ s

**(1 mark)**

**5** A physical quantity has the unit m $s^{-2}$ and the symbol $a$.

What is the correct name for this quantity?

**A.** Activity of a radioactive source

**B.** Amplitude

**C.** Avogadro constant

**D.** Acceleration

**(1 mark)**

**6** Which of the following must all physical quantities have?

**A.** Direction and magnitude

**B.** Direction and a unit

**C.** Magnitude and a prefix

**D.** Magnitude and a unit

**(1 mark)**


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# Medium Questions

**1** A student creates a table to show reasonable estimates of some physical quantities.

Which row is a reasonable estimate?

| | quantity | value |
|---|---|---|
| **A** | frequency of a radio wave | 500 pHz |
| **B** | mass of an atom | 500 μg |
| **C** | the Young modulus of a metal | 500 kPa |
| **D** | wavelength of green light | 500 nm |

**(1 mark)**

**2** What is a reasonable estimate for the volume of 1 ml of water?

**A.** 0.1 $cm^3$

**B.** 1 $cm^3$

**C.** 10 $cm^3$

**D.** 100 $cm^3$

**(1 mark)**

**3** Which row shows what all physical quantities must have?


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| | magnitude | direction | unit |
|---|---|---|---|
| **A** | ✗ | ✗ | ✓ |
| **B** | ✓ | ✗ | ✓ |
| **C** | ✓ | ✓ | ✗ |
| **D** | ✓ | ✓ | ✓ |

**(1 mark)**

**4** An adult male stands with a litre carton of milk balanced on his head.

What is the best estimate of the gravitational potential energy of the milk carton?

**A.** 2 J

**B.** 20 J

**C.** 200 J

**D.** 2000 J

**(1 mark)**

**5** Which of the following is a reasonable estimate for the density of a sugar cube?

**A.** 0.01 kg $m^{-3}$

**B.** 0.1 kg $m^{-3}$

**C.** 10 kg $m^{-3}$

**D.** 1000 kg $m^{-3}$

**(1 mark)**


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# Hard Questions

**1** Which row shows a reasonable estimate?

| | quantity | value |
|---|---|---|
| **A** | kinetic energy of a bus travelling on an expressway | 30 000 J |
| **B** | power of a domestic light bulb | 300 W |
| **C** | temperature of a hot oven | 300 K |
| **D** | volume of air in a car tyre | 0.03 $m^3$ |

**(1 mark)**

**2** Which statement is **incorrect** by a factor of 100 or more?

**A.** Atmospheric pressure is about $1 \times 10^5$ Pa

**B.** Light takes $5 \times 10^2$ s to reach us from the Sun

**C.** The frequency of ultraviolet light is $3 \times 10^{12}$ Hz

**D.** The life-span of a man is about $2 \times 10^9$ s

**(1 mark)**


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**3** An adult person jumps from a wall which is 2 m high. They land without rebounding.

What is the best estimate for the change in momentum of the person?

**A.** $4 \times 10^{0}$ kg m $s^{-1}$

**B.** $4 \times 10^{1}$ kg m $s^{-1}$

**C.** $4 \times 10^{2}$ kg m $s^{-1}$

**D.** $4 \times 10^{3}$ kg m $s^{-1}$

**(1 mark)**

**4** The diameter of a proton is approximately $1 \times 10^{-15}$ m.

Which estimate is reasonable for the time taken for light to cross the nucleus of a hydrogen atom?

**A.** $3 \times 10^{-28}$ s

**B.** $3 \times 10^{-24}$ s

**C.** $3 \times 10^{-20}$ s

**D.** $3 \times 10^{-16}$ s

**(1 mark)**

**5** The bottom of the Mariana Trench is estimated to be approximately 11 000 m below sea level.

What is the best estimate of the total pressure at the bottom of the Mariana Trench?

**A.** $1 \times 10^{6}$ Pa

**B.** $1 \times 10^{7}$ Pa

**C.** $1 \times 10^{8}$ Pa

**D.** $1 \times 10^{9}$ Pa

**(1 mark)**


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**6** A satellite orbits Earth at an altitude of 400 km above the Earth's surface.

What is a reasonable estimate of the total distance travelled by the satellite in one complete orbit?

**A.** $4 \times 10^5$ m

**B.** $4 \times 10^6$ m

**C.** $4 \times 10^7$ m

**D.** $4 \times 10^8$ m

**(1 mark)**


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